CENTURION UNIVERSITY • B.TECH CSE 7TH SEMESTER

Micro-ROS Enabled Autonomous Environmental Rover

Subject: Robotic ROS with C++

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Team: Jagannath Mohapatra, Khawar Ahemed Khan, Amarendra Das, Polikey Bhuvan

Project Overview: What We Are Building

The Core Concept

Building an embedded telemetry rover by pushing the powerful ROS 2 ecosystem onto an ultra-cheap ESP32 microcontroller for under $30.

Primary Capabilities

Continuous ambient thermal mapping over Wi-Fi, paired with zero-latency hardware collision avoidance via ultrasonic arrays.

Real-World Application

Designed for scalable mass deployment in warehouse security patrols, smart agriculture micro-climate tracking, and hazardous search and rescue.

Engineering Roster & Current State

State: Finalizing Connection Tests Before Coding

230301120151
Jagannath Mohapatra
ESP32 MCU Assembly

230301120121
Khawar Ahemed Khan
Sensor Integration

230301120146
Polikey Bhuvan
Motors & Tires Assembly

230301120137
Amarendra Das
Connection Testing

Jagannath: ESP32 Microcontroller Assembly

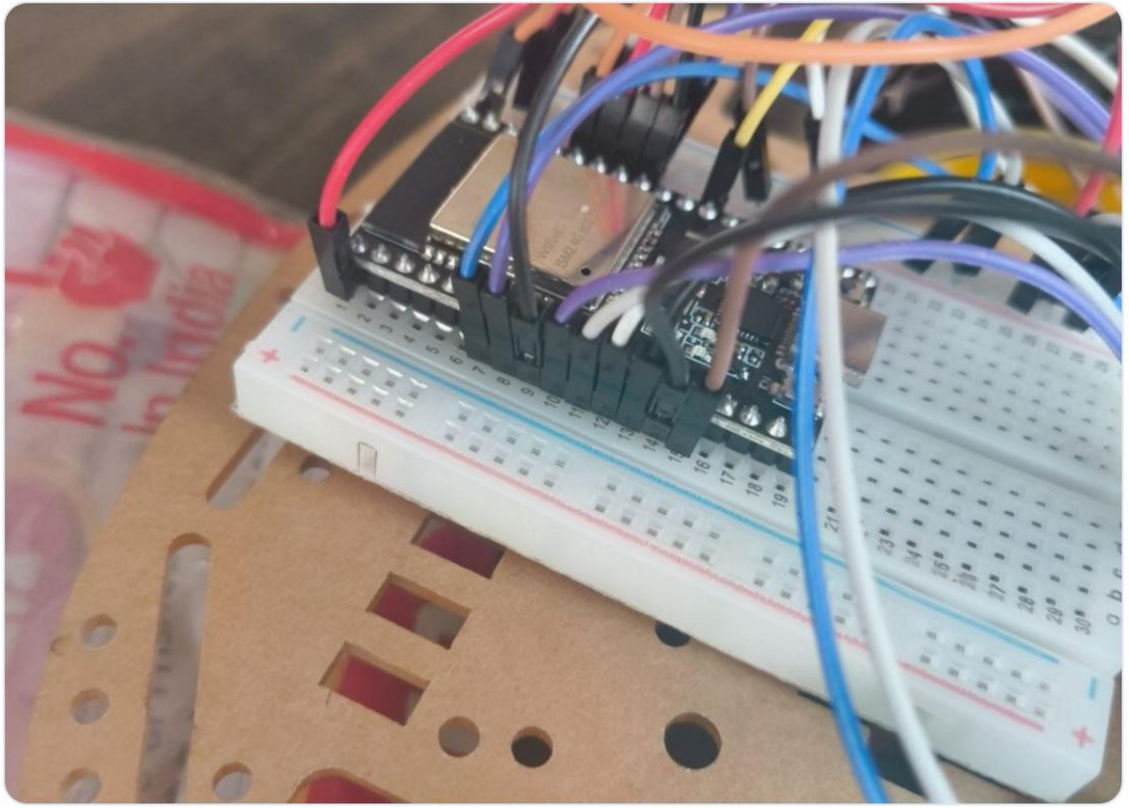
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- **Chassis integration:** Secured the breadboard to the acrylic base.
- **Mounted core logic:** Installed the 38-pin ESP32 dev board.
- **Pinout mapping:** Aligned structural pins for later serial communication.



Jagannath: Why It Matters (Logic Trace Isolation)

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What was done

Routed isolated jumper traces across the breadboard. Separated logic rails from motor power distribution.

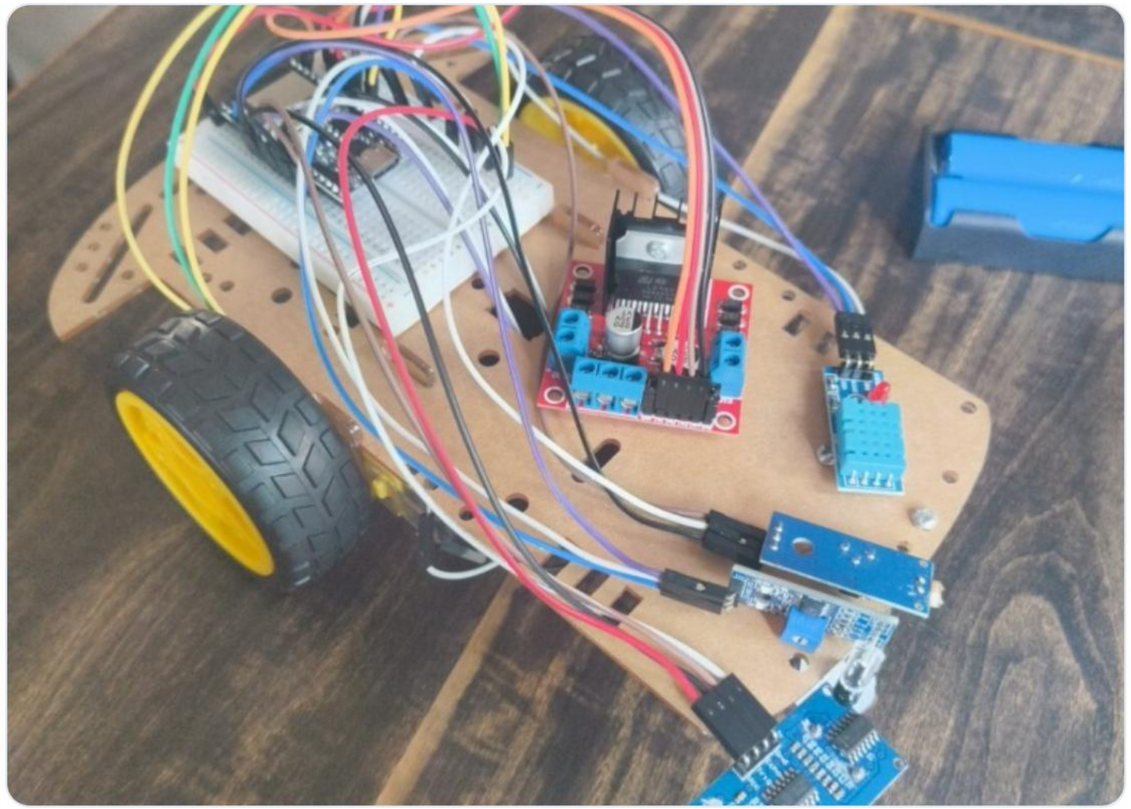
Why it matters

Ensures clean 5V logic power to the ESP32. Prevents catastrophic MCU brownouts and reboots during high-current motor spikes.

Khawar: Sensor Array Integration

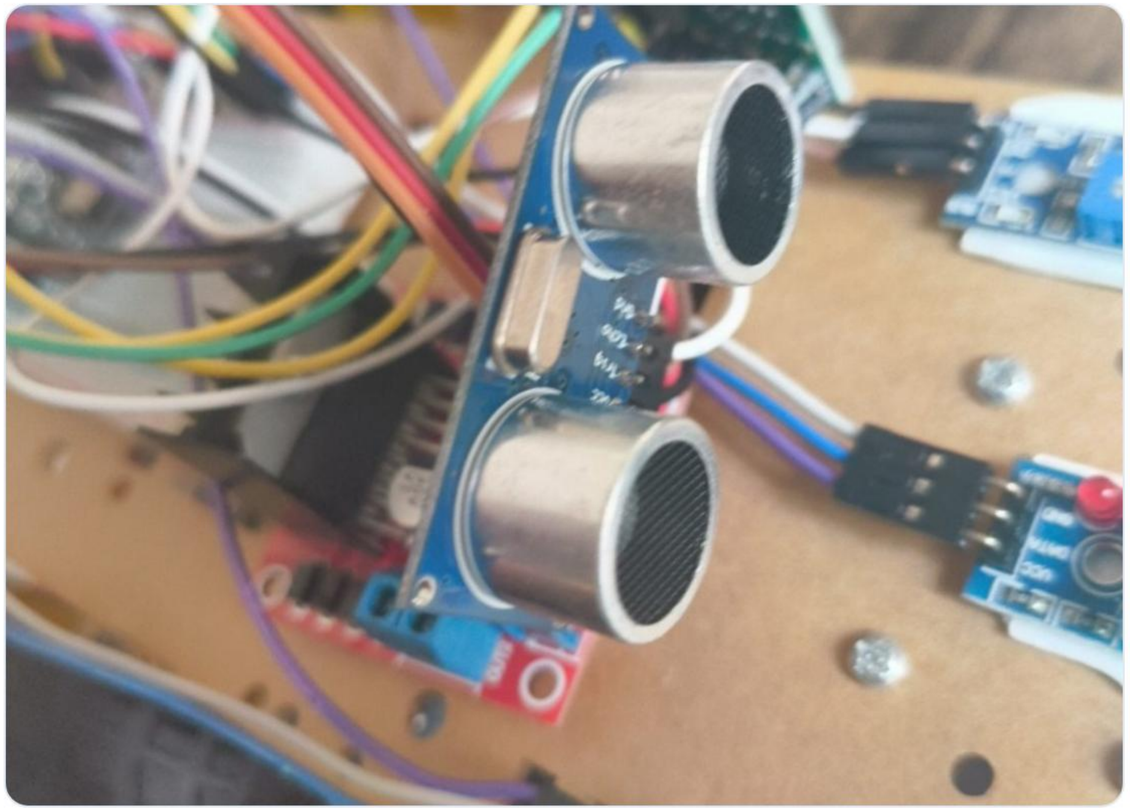
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- **Environmental payload:** Assembled DHT11 temperature and humidity units.
- **Spatial arrays:** Mounted Ultrasonic and IR modules.
- **Data routing:** Connected sensor pins to the MCU logic rails.



Khawar: Why It Matters (FOV Placement Strategy)

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What was done

Positioned ultrasonic and IR sensors strictly on the forward edge. Elevated environmental sensors away from heat sources.

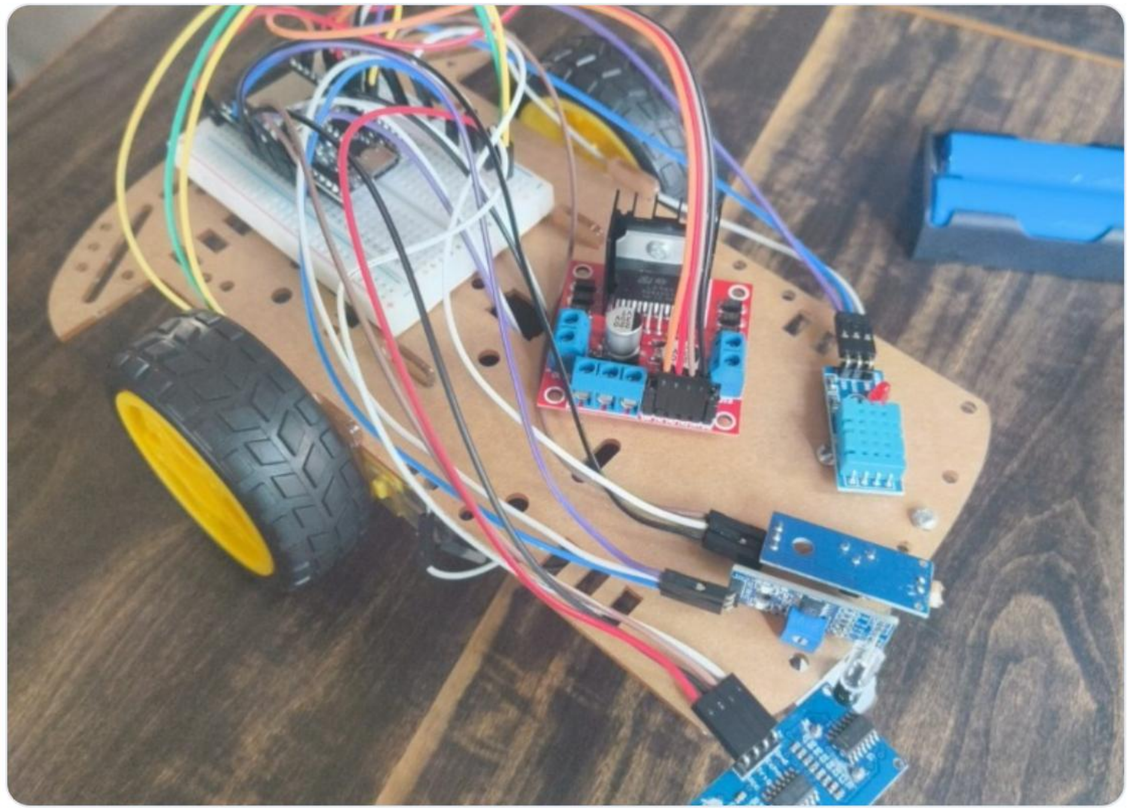
Why it matters

Secures an unobstructed forward field-of-view. This physical placement is absolutely critical for zero-latency hardware collision avoidance to function.

Bhuvan: Robot Tires & Motors Assembly

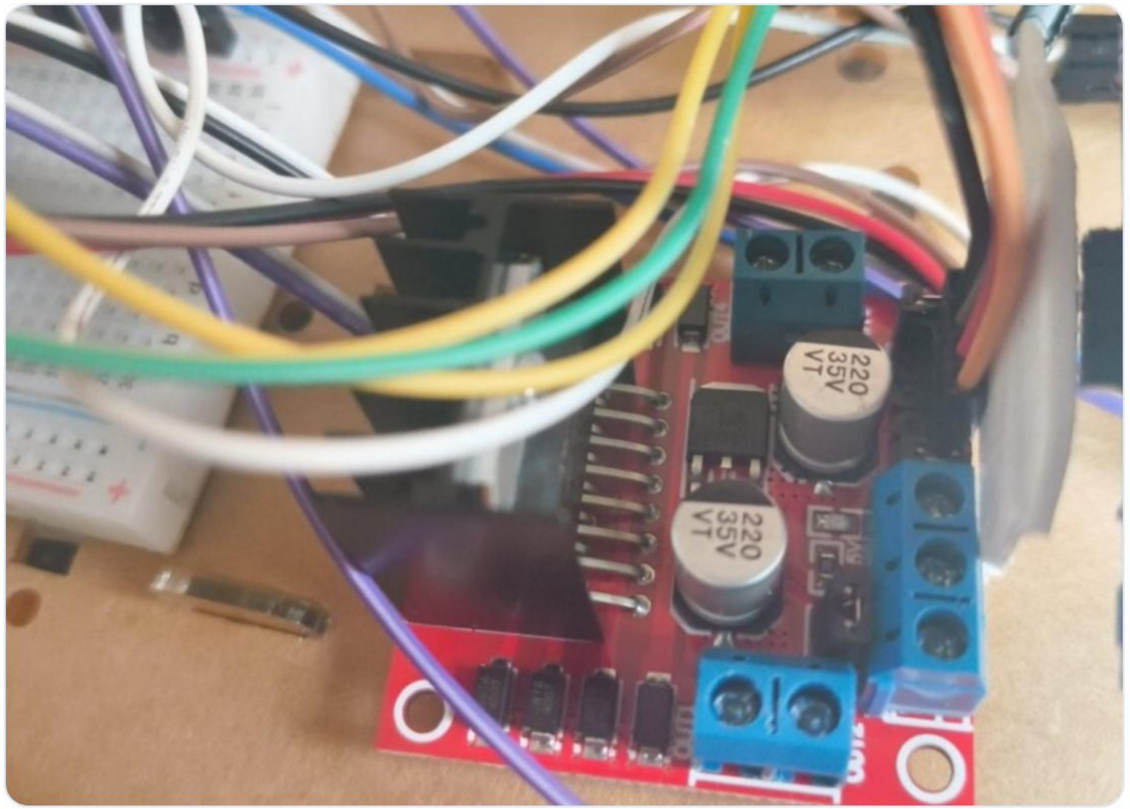
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- **Chassis build:** Fixed and aligned the acrylic platform structures.
- **Actuation:** Anchored the dual BO motors securely.
- **Traction:** Secured rubberized wheels to handle uneven terrain.



Bhuvan: Why It Matters (High-Current Power Routing)

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What was done

Integrated the red L298N motor driver module. Routed direct 7.4V lines from battery packs to the motor terminals.

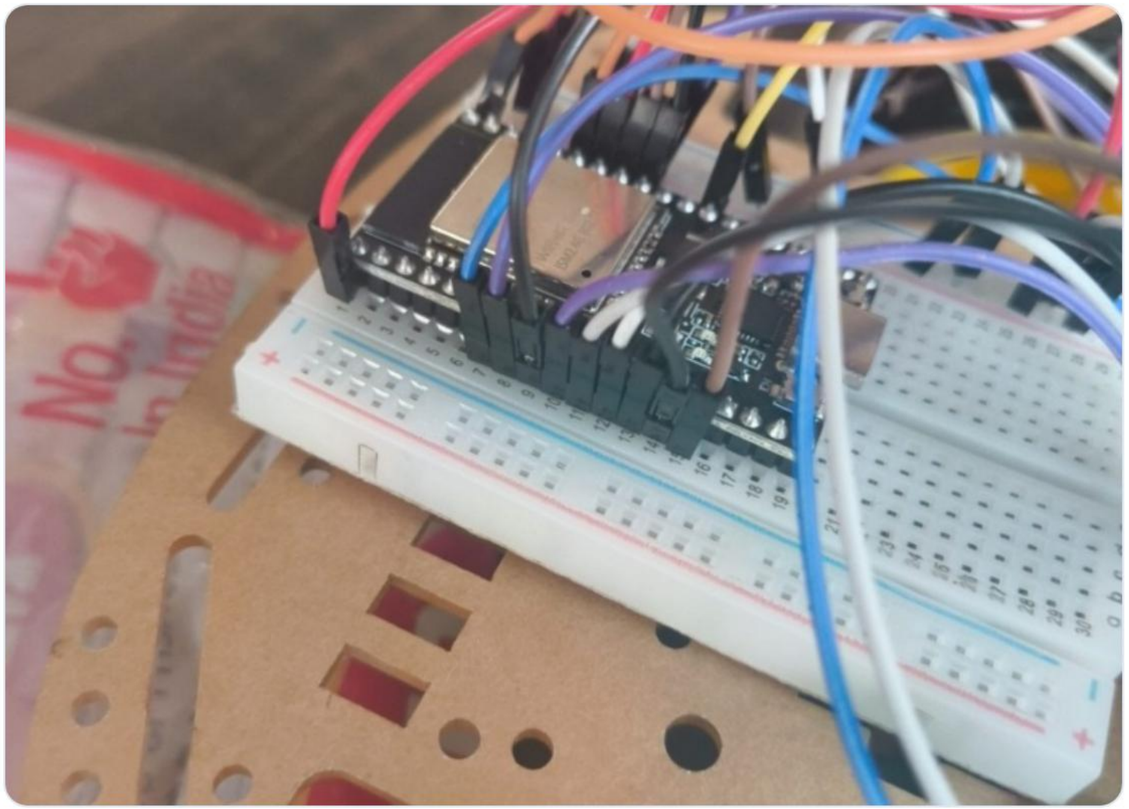
Why it matters

Distributes high-current capacity safely. This specific module is necessary to translate the ESP32's low-voltage PWM signals into raw mechanical torque.

Amarendra: Connection Testing

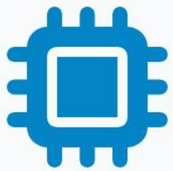
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- **Circuit verification:** Tested all 60+ jumper connections on the breadboard.
- **Voltage checks:** Inspected power rails for proper isolation.
- **Short prevention:** Secured exposed terminal blocks before power-on.



Amarendra: Why It Matters (Firmware Readiness)

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100% Hardware Ready

What was done

Finalized physical layer validation. Transitioning the project state from hardware building to software flashing.

Why it matters

Prevents electrical failures and short circuits during code execution. Prepares the board securely for the upcoming C++ Micro-ROS flashing.

Next Phase: Software Architecture & Programming

Micro-XRCE-DDS Bridge

Standard ROS is too heavy. We will configure XRCE middleware to bridge the ESP32 directly to Ubuntu via Wi-Fi.

Embedded C++ & Polling

Using rclc logic to establish node identity. A 100ms hardware timer will sample DHT11 pins without blocking other processes.

Actuation Override

Programming C++ failsafes. The MCU will instantly drop PWM to zero if ultrasonic detects objects under 10cm, ignoring network lag.

Next Steps & Future Scope

IMMEDIATE

Firmware Flash

Deploy the C++ code to the ESP32. Establish live Wi-Fi telemetry with the Ubuntu Master dashboard.

PHASE 2

LiDAR Integration

Upgrade standard IR sensors to a 2D LiDAR module. Enables full 360-degree room mapping.

PHASE 3

SLAM Navigation

Implement Simultaneous Localization and Mapping. Unlocks true autonomous pathfinding via ROS 2.



Project Conclusion

Combining affordable embedded hardware with the powerful ROS 2 ecosystem creates a scalable, accessible tool for advanced environmental robotics.

Thank You • Questions & Live Defense Welcome